FORMULAE LIST

Volume of a sphere: $V = \frac{4}{3}\pi r^3$ Volume of a cone: $V = \frac{1}{3}\pi r^2 h$ Volume of a pyramid: $V = \frac{1}{3}Ah$

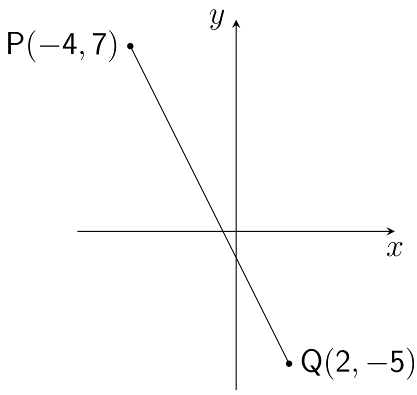
1. Evaluate $3\frac{3}{4} \div \frac{6}{7}$. (2)

2. Solve $x^2 + 8x - 20 = 0$. (2)

3. Simplify $\dfrac{w^4}{w^2 \times w^{-5}}$. (2)

4. Solve the equation $2(4x - 3) - 5(x - 1) = 6$ (3)

5. The diagram shows a straight line connecting points P and Q. (3)



Find the equation of line PQ. Give the equation in its simplest form.

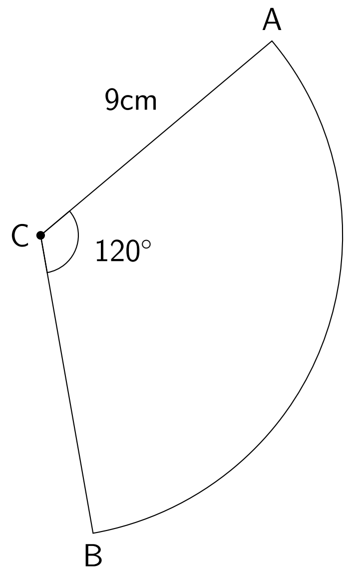
6. Expand and simplify $(2x - 3)(x^2 - 5x + 4)$. (3)

7. The tail lengths of a sample of nine red squirrels were recorded, in centimetres, as follows: (3)

9 10 12 12 13 13 15 15 19

Find the median and interquartile range of these values.

8. The diagram below shows a sector of a circle, centre C, with radius 9 centimetres. (3)



Angle ACB $= 120°$.

Calculate the length of arc AB.

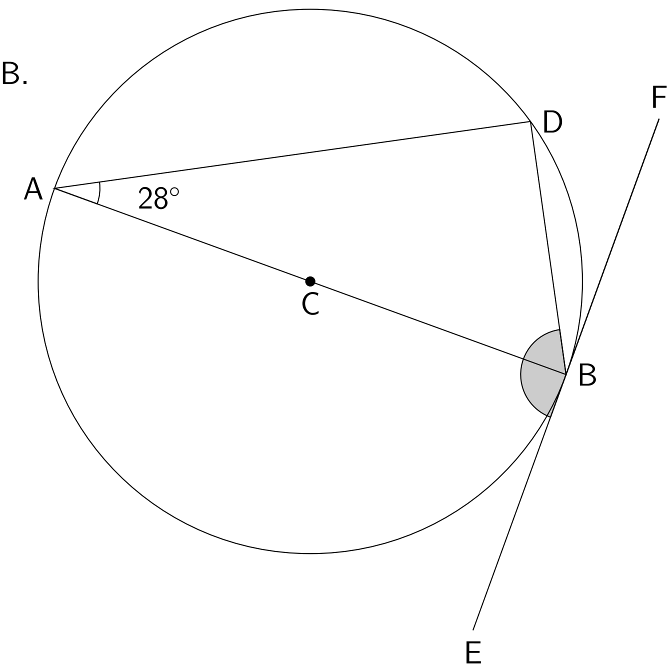
Use $\pi = 3.14$.

9. Solve, algebraically, the equation $\dfrac{3x+2}{5} + 1 = 2x - 7$. (3)

10. Factorise fully $3y^2 - 48$. (2)

11. In the diagram below, which contains the circle with centre C: (3)

- Line AB passes through C.
- EF is a tangent to the circle at B.
- Angle DAC $= 28°$.



Find the size of angle DBE.

12. The straight line passing through $(-4, 1)$ and $(5, k)$ has a gradient of $\dfrac{2}{3}$. (2)
Find the value of k.

END OF PRACTICE EXAM A PAPER 1

HINTS: Practice Exam A Paper 1

1. To divide by a fraction, it can be rewritten as a **multiplication**...
2. To solve a **quadratic equation** which is arranged "$= 0$", try **factorising**.
3. When **multiplying** powers, we **add** the indices. When **dividing**...
4. When multiplying out the brackets, take care when multiplying by **negative 5**.
5. Starting by calculating the **gradient** between the coordinates.
6. Check your powers and **negatives** carefully.
7. To find the **median**, first check the values are in order...
8. An **arc length** is a **fraction** of the **circumference**.
9. To deal with the fraction, multiply through by the **denominator**...
10. Start with the **common factor** then look inside the brackets to see what *else* can be done.
11. Start by looking for any **right-angles**...
12. What formula connects **two coordinates** and the **gradient** of the line connecting them?

ANSWERS: Practice Exam A Paper 1

1. $\frac{35}{8}$ **or** $4\frac{3}{8}$
2. $x = -10, x = 2$
3. w^7
4. $\frac{7}{3}$
5. $y = -2x - 1$
6. $2x^3 - 13x^2 + 23x - 12$
7. median $= 13$, IQR $= 4$
8. 18.84cm
9. $x < 6$ **or** $6 > x$
10. $3(y + 4)(y - 4)$
11. Angle DBE = 152°
12. $k = 7$